TAMU 2025 ECEN 607 Advanced Analog Circuit Design Project

100 MHz pipeline data converter with IBM-180nm

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I. *Introduction*

This project presents a high-performance 100 MHz pipeline Analog-to-Digital Converter (ADC) with 11-bit resolution and 2.5 bits per stage. It targets a Signal-to-Noise Ratio (SNR) of approximately 67.7 dB SNR. Key transistor-level designs include comparators with offsets under 20 mV and fully differential residue amplifiers for the initial two stages, optimized for accuracy and speed. Advanced bootstrap switches are utilized to minimize charge injection and clock feedthrough, ensuring linearity and dynamic range.

II. *DESIGN PROCEDURE*

Figure 1 illustrates a typical pipeline ADC, consisting of a first-stage Track & Hold followed by an MDAC stage (Flash ADC and residue amplifier), along with latch circuits that delay the digital output sum.

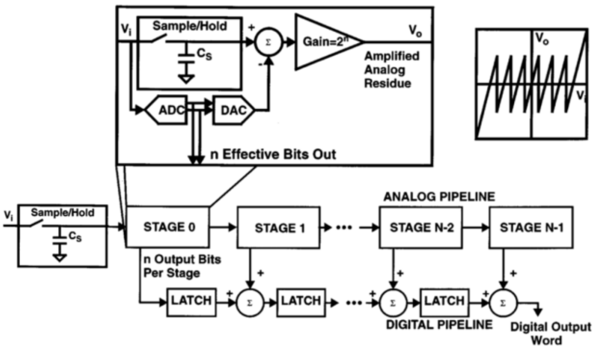


Figure 1. typical pipeline ADC

■ *2.5-bits ADC*

Figure 2 illustrates the concept of a 2.5-bit pipeline ADC compared to a 3-bit ADC. A 2.5-bit ADC achieves nearly the same resolution as a 3-bit ADC but uses fewer comparators. Although the theoretical maximum residue amplifier gain is 2^N=8 for a 3-bit stage, setting the gain at 2^{N-1}=4 for a 3-bit stage increases tolerance to comparator offset and amplifier gain errors. However, this choice sacrifices the input range near the extremes (±7/8Vref to ±Vref), slightly reducing the usable input voltage range in exchange for robustness and lower complexity.

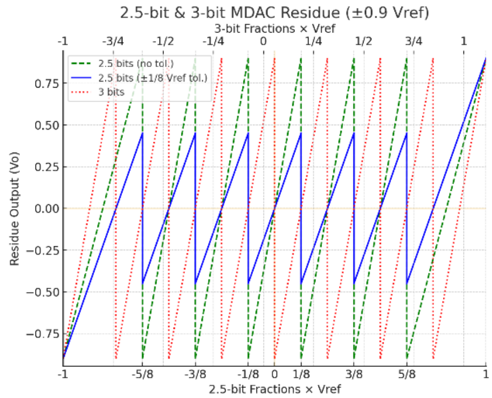


Figure 2. 2.5-bit ADC V.S. 3-bit ADC, Vi V.S. VO

2.5-bit ADC: Gain=4, less comparators, tolerating offset

3-bit ADC: No matter if Gain=4 or 8 (Gain=8 can't tolerate offset) still need more comparators

2.5-bit ADC no tolerance: Gain=8, can't tolerate any offset

Because every MDAC is designed for a gain of 4 (2^2), the 3-bit word produced by one stage does **not** all pass straight to the output: its least-significant bit overlaps the next stage's most-significant bit. In the digital domain this means "3 bits + 3 bits → 5 effective bits," or, equivalently, a digital left-shift of two places (gain = 4).

If we instead used the maximum residue gain of 8, each stage would need to deliver a perfectly accurate 6-bit result (3 bits from the current stage plus 3 bits from the next). With gain = 8 the operational amplifier can tolerate essentially **no settling or offset error**, so the relaxed-gain-4 architecture is preferred: it trades one bit of output range for a comfortable error margin.

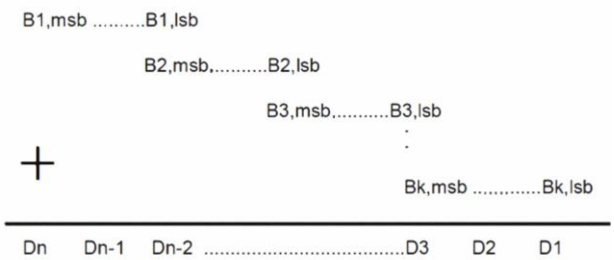


Figure 3. Output Digital Form

■ Flash ADC

In a Flash ADC, the input voltage Vin is simultaneously compared against a ladder of reference levels Vref, , with each comparator outputting “1” if Vin>Vref, (and “0” otherwise), forming a thermometer code. A priority encoder then locates the highest-order “1” and emits its index as an NNN-bit binary word, achieving single-cycle conversion with minimal logic depth and ultra-low latency

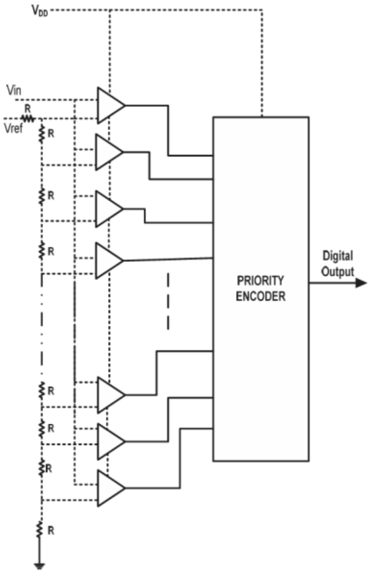


Figure 4. N bit-Flash ADC

However, beyond about 5 bits of resolution, the exponential growth in comparator count and the cumulative effect of comparator mismatches make Flash architectures increasingly error-prone and impractical.

■ Flash ADC encoder

In a Flash ADC, the comparator array produces a thermometer code—contiguous “1”s up to the threshold point. A priority encoder then finds the highest order “1” and outputs its index as a binary word, converting thermometer to binary with minimal logic and low latency.

Input Signal	Priority Encoder		
Vin (-0.9~0.9)	X2	X1	X0
0.562<Vin<0.787	1	1	0
0.337<Vin<0.562	1	0	1
0.112<Vin<0.337	1	0	0
-0.112<Vin<0.112	0	1	1
-0.337<Vin<-0.112	0	1	0
-0.526<Vin<-0.337	0	0	1
-0.787<Vin<-0.526	0	0	0

Figure 5. 2.5bit Flash ADC In-Out Table

■ Comparator

A dynamic comparator works in two clocked phases: on CLK low it resets the internal nodes and “samples” the small in+/in– voltage difference onto parasitic capacitances; on CLK high it turns on the cross-coupled inverters, whose positive feedback rapidly regenerates that tiny difference into full-swing digital outputs in one cycle.

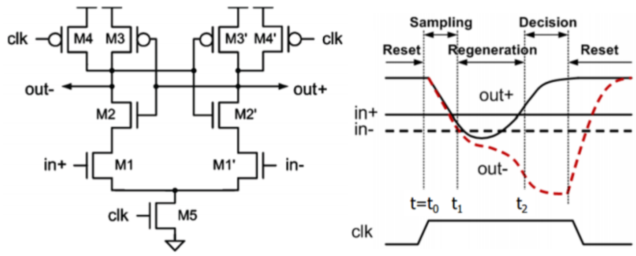


Figure 6. Strong Arm Operation

An SR latch placed after the comparator serves to latch and hold the comparison result until the next clock cycle, isolating the comparator’s dynamic output from downstream logic. It prevents glitches and race conditions by providing a stable, synchronized digital output for subsequent stages.

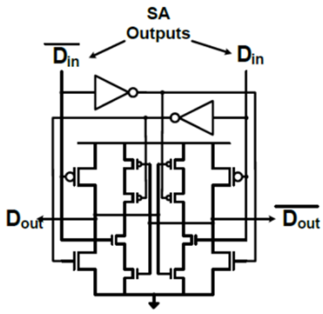


Figure 7. SR latch

■ Residue Amplifier

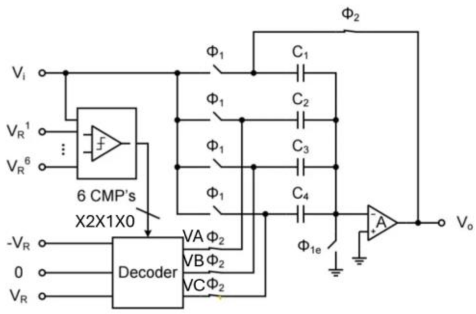


Figure 8. 2.5bit MDAC

In each pipeline stage, the input voltage Vin is first sampled onto a switched-capacitor array and a small flash ADC produces a coarse digital decision D. A DAC then injects D×Vref back onto the same caps, subtracting the estimated level and leaving a residue ΔV=Vin−D Vref. Finally, a closed-loop amplifier with gain G re-scales this residue (so Vout=G × ΔV to fill the full dynamic range of the next stage.

By amplifying each residue to full-scale, the quantization error is shifted into the next stage's range—letting each stage resolve an extra "bit"—and cascading these stages delivers the pipeline ADC's high overall accuracy.

By charge conservation at the summing node ($\varphi_1 \rightarrow \varphi_2$), the total charge on the four-capacitor array must remain constant:

4C1 Vin(ϕ1) = C1 Va+C1 Vb+C1 Vc+C1 Vo (ϕ2)

Divide through by C1 and rearrange for Vo :

Vo = 4 Vin − (Va+Vb+Vc) = 4 (Vin−Va/4+Vb/4+Vc/4).

Thus, the residue amplifier both subtracts the DAC-injected sum (Va+Vb+Vc) /4 from the sampled input and multiplies the result by the stage gain G=4.

Vi	Thermal e6e5e4 e3e2e1	Encode X2X1X0	DAC Level	Va Vb Vc	Vo
Vi < -0.562	000000	000	-0.675	-1,-1,-1	4(vi+0.675)
-0.562 < Vi < -0.337	000001	001	-0.45	-1,-1,0	4(vi+0.45)
-0.337 < Vi < -0.112	000011	010	-0.225	-1,0,-0	4(vi+0.225)
-0.112 < Vi < 0.112	000111	011	0	0,0,0	4(vi)
0.112 < Vi < 0.337	001111	100	0.225	1,0,0	4(vi-0.225)
0.337 < Vi < 0.562	011111	101	0.45	1,1,0	4(vi-0.45)
Vi > 0.562	111111	110	0.675	1,1,1	4(vi-0.675)

Figure 9. Residue Amplifier table

The diagram below illustrates how the residue amplifier operates during φ_1 and φ_2 in the five-stage pipeline ADC.

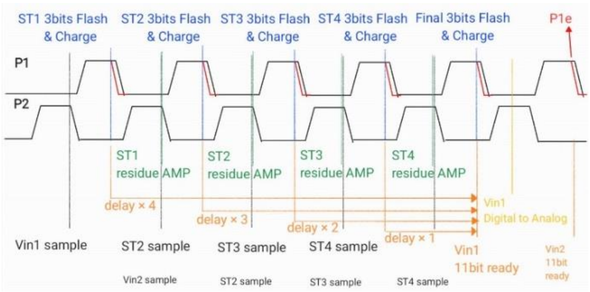


Figure 10. Operation clock in each stage MDAC

■ Residue OPA

In this project we used a dual-input two-stage amplifier (D2SA): the first cascode stage delivers a large DC gain and high output impedance, while the second Class-AB stage provide large slew rates and drive heavy capacitive loads. The fully differential dual-input front end boosts CMRR and halves input capacitance, with a common-mode feedback loop stabilizing the output common-mode voltage. D2SA achieve wide GBW and ample phase margin, resulting in a compact, low-power amplifier with high accuracy, fast settling, and strong drive capability. (Figure11)

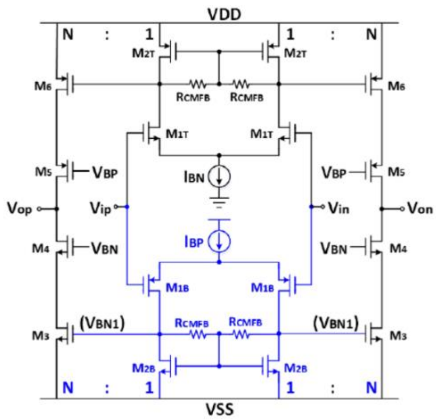


Figure 11. D2SA

Opa finite **gain** introduces error to the next stage in the pipelined ADC. To achieve the accuracy requirement for an N-bit pipelined ADC, the gain error contributed by the first stage with M-bit resolution (less than LSB $1.8/2^{11}$) should satisfy the equation

$$\left(\frac{1}{\beta}-\frac{A}{1+A\beta}\right)\frac{V_{ref}}{2^{M}} \le \frac{V_{ref}}{2^{N-M}} \qquad \left(\frac{1}{B}= 4 \text{ for 2.5-bit stage}\right)$$

Equation can be simplified to: A >= 2^{N-11} For 11 bits ADC, minimum A is 60dB, choosing **70 dB** for process variation and mismatch errors.

With the same **GBW** product, the same amplifier, if placed in a negative feedback system to achieve a gain of 4 will have a bandwidth equal to

$$GBW = gain \times bandwidth = 4 \times bandwidth \Rightarrow bandwidth = \frac{GBW}{4}$$

For an N-bit pipeline ADC, the op-amp's unity-gain bandwidth must greatly exceed the sampling rate so that bandwidth-induced error stays below 1 LSB of the remaining-bits resolution.

$$\left[\frac{1}{\beta}-\frac{1}{\beta}\times\left(1-e^{\frac{-t}{\tau}}\right)\right]\frac{V_{ref}}{2^{M}}\le\frac{V_{ref}}{2^{N-M}}$$

1/B stands for inter-stage gain, Vref is the reference voltage of ADC, $1\text{-}e^{-t/\tau}$ is the step response of feedback system with τ=1/2π*B*fu, fu= unity frequency.

$f_u > \frac{(N-M)\ln(2)}{2\pi.\beta.T_{settling}}$ T settling= $\frac{1}{200M*}*\frac{1}{2}*\frac{8}{10}$= 2ns, **fu= 1.76G Hz**

Slew rate is the fastest speed the output can change—set by the drive current and load capacitance—and a low value causes distortion near full-scale swings.

$$SR > \frac{dV_{\max}}{dt} = \frac{Voltage_{swing-Diff}\times 2\pi}{T_{settling}}$$

Select the sampling **capacitor** so that its kT/C thermal noise power is kept below the equivalent of one LSB.

kT/C= Δ^2 /12, C= 12kT * (2^N / VFS)^2

C=65f F for 11bits

■ Bootstrapped Switch

In a bootstrapped sampler, the MOS sampling switch's gate is "bootstrapped" to its source via a dedicated capacitor, keeping VGS constant and thus flattening the on-resistance over the entire input swing. During the **track** phase (φ_1), isolation switches tie the bootstrap capacitor and sampling transistor together, so Vin is stored on C1; in the **hold** phase (φ_2), the bootstrap capacitor is disconnected, recharged to VDD , and the sampled charge is held for downstream conversion. This two-phase scheme minimizes switch distortion, enables multi-GHz operation with minimal power, and yields clean, low-error sampling essential for high-speed Nyquist ADCs (Figure 1).

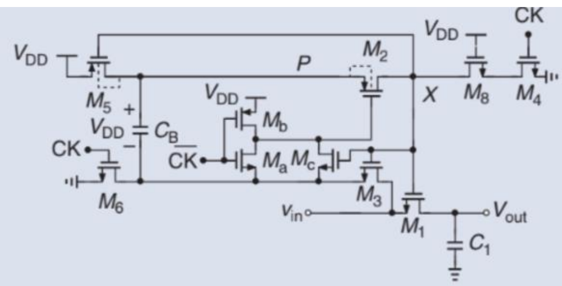


Figure 12. Bootstrapped Switch

III. Schematic & simulation results

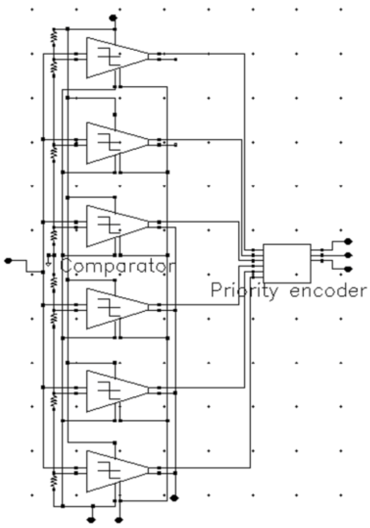


Figure 13. 2.5bit Flash ADC

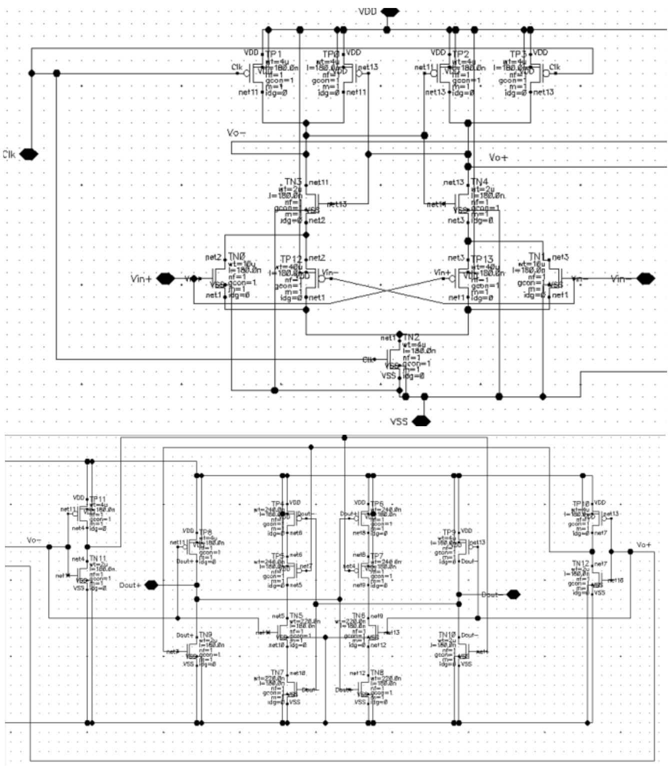


Figure 14. StrongArm connecting to SR Latch

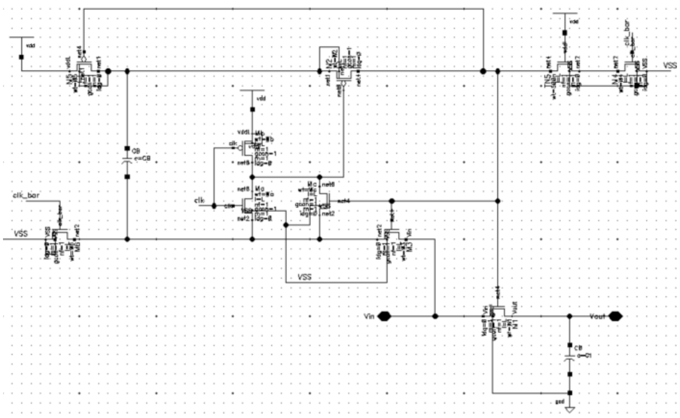


Figure 15. Bootstrapped switch circuit

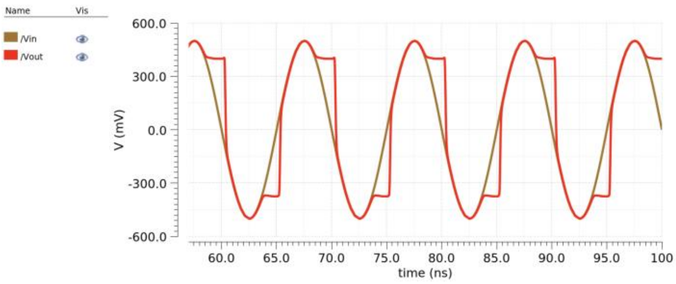


Figure 16. Simulation for Bootstrapped switch

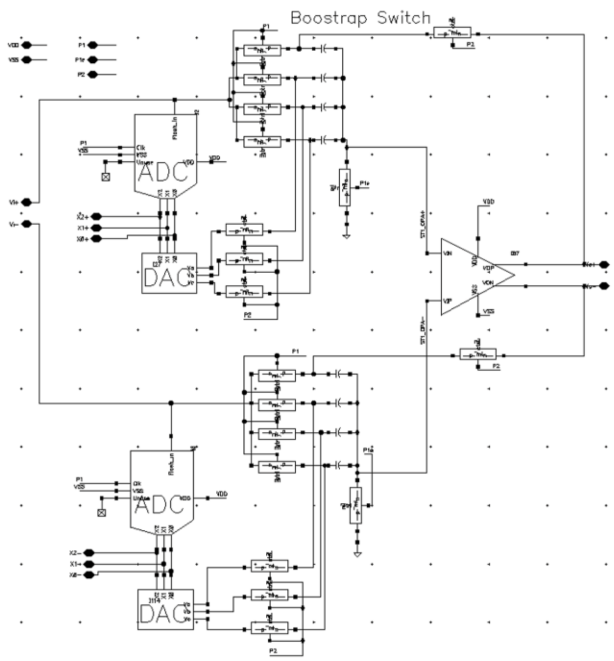


Figure 17. StageN MDAC with residue amplifier

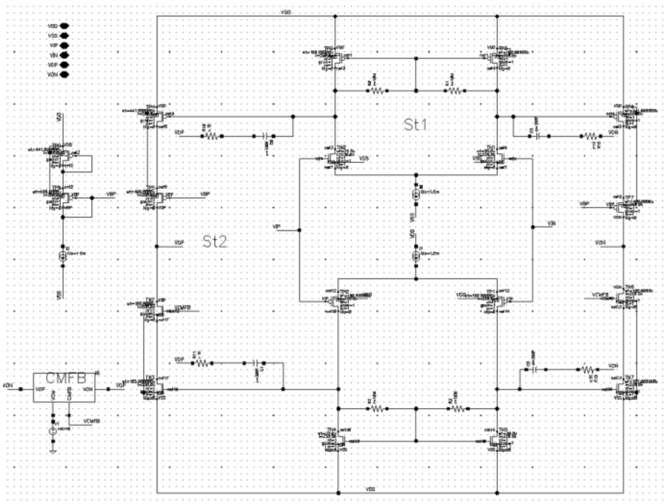


Figure 18. D2SA OPA

Figure 19. Open loop Gain and GBW

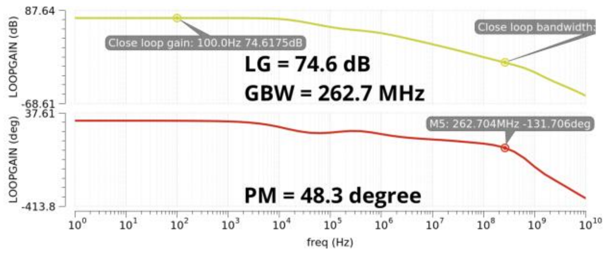


Figure 20. Close loop Gain, GBW and PM under φ_2 (B=$\frac{1}{4}$)

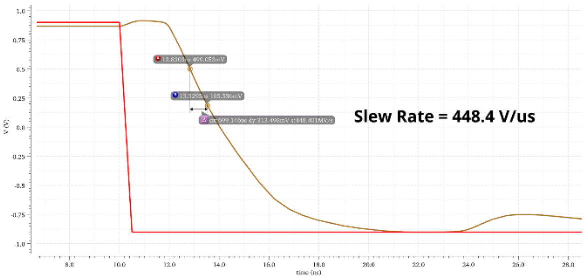


Figure 21. Measured slew-rate

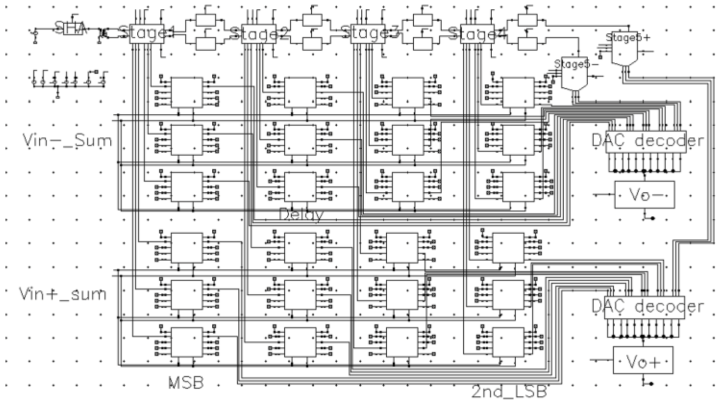
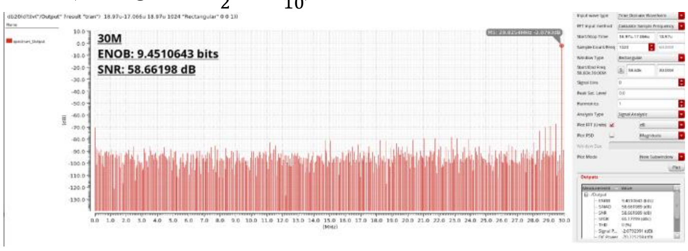
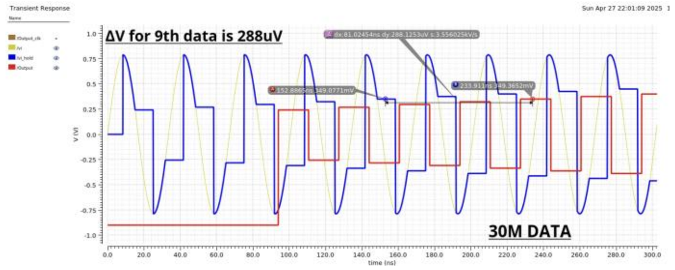


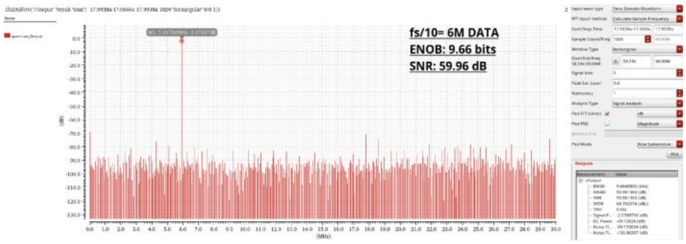
Figure 22. Stage (2.5 per stage) Pipeline ADC

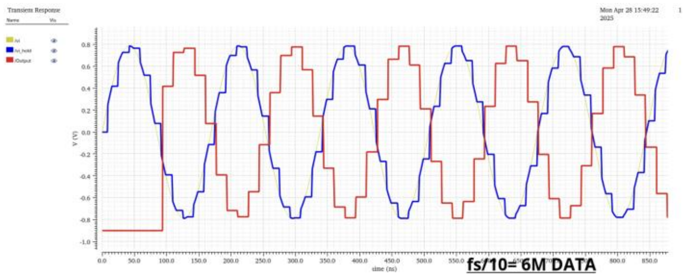
When selecting the test frequency, we set fin=N Fs/M. M is chosen as a power-of-two so the FFT bins line up neatly, while the DFT length Nis made prime; this prevents the sampled points from repeating in a short cycle and suppresses spurious periodic artefacts. Because a prime N guarantees the input period is non-integer with respect to the record length, any residual spectral leakage is easily confined to neighboring bins by a mild taper (e.g., a Hann window), preserving amplitude accuracy. Finally, we keep Fs≥2fin to satisfy Nyquist and avoid aliasing of the highest test tone.

- Using 30M data with 60M sampling clock testing (testing with $\frac{fs}{2}$ & $\frac{fs}{10}$)

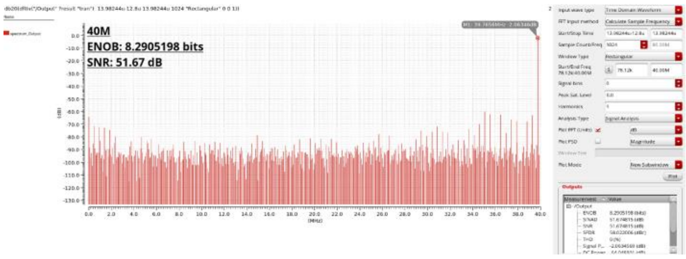


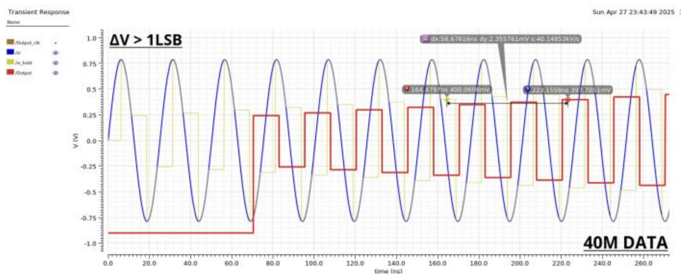




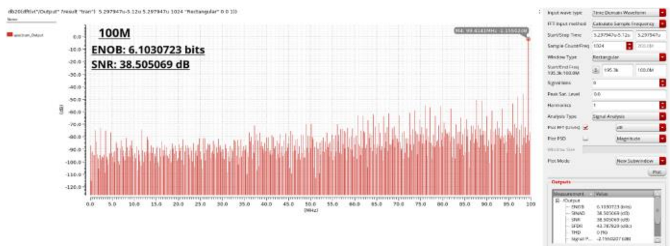


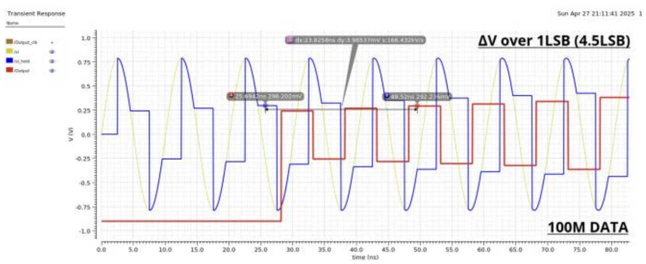
➢ Using 40M data with 80M sampling clock testing





➢ Using 100M data with 200M sampling clock testing





IV. Conclusion

Sample frequency	60 MHz	80 MHz	200 MHz
Data frequency	30 MHz	40 MHz	100 MHz
ENOB (bits)	9.66	8.29	6.1
SNR (dB)	59.96	51.67	38.5

We aim for a 100 MHz pipeline ADC at IBM 180 nm. Although our calculated slew-rate and GBW targets indicate feasibility, in practice they leave little design margin. Because SR and GBW set each stage's settling accuracy, limited values can introduce residual error—an effect that grows as the sampling clock increases and the settling window shrinks. A brief literature survey suggests that many 180 nm pipeline-ADC implementations operate around 30–40 MHz, highlighting the additional care required when pushing toward 100 MHz rather than proving it unattainable ([6] [7]).

V. Reference

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